

Global Facilities - Design & Construction - Construction - Cleanroom Protocol

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1.0 Purpose

- 1.1 This document provides the standards and guidelines concerning construction of new areas to be used as cleanroom facilities as well as the requirements for cleanroom certification.

2.0 Overview

- 2.1 Micron operates a strict program to ensure successful construction of a new or remodeled cleanroom area.
- 2.2 There are five separate levels in this program, and each level has three separate categories.

- Construction Activities: Describes what type of construction is taking place.
- Cleanroom Protocol: Describes what type of procedures and cleanroom protocol must be followed in order to maintain total cleanliness in the particular area under construction.
- Cleanroom Gowning: The type of garments to be worn will be determined by the current stage of construction in that given area.

Important: Every cleanroom area being constructed requires some type of garments to be worn. These garments are to protect the environment. They must be kept clean and

not used for overalls or aprons. All garment protocols will be enforced.

- 2.3 The five stages detailed in [Section 6.0](#) were created to inform the building supplier and contracted workforce of the cleanroom protocol and expected construction progress at each stage.
- 2.4 During cleanroom construction, it is likely that there will be several areas within the superstructure with different cleanliness levels and protocols. It is important to ensure that these different zones/stages of construction are well segregated and that protocol for each zone/stage is strictly followed and enforced.
- 2.5 Every effort will be made to keep emergency egress exits clear of obstructions. If there is a need to block an existing exit; temporary evacuation maps will be placed in the area designating the nearest alternate exit.

3.0 Definitions

- **SOP** - Standard Operating Procedure
- **Initial Certification** - Completed once the "As-Built" status is achieved. Testing is to demonstrate that airborne and surface particle counts, temperature and humidity, surface resistance, and air balance and flow are within project specifications. Testing is also to confirm that the ceiling grid and ULPA/HEPA filters are sealed and functioning as designed. See [Section 6.7.4](#) for more details.
- **Intermediate Certification** - Completed once the "At-Rest" status is achieved. Testing is to demonstrate that airborne and surface particle counts, and temperature and humidity are within project specifications. See [Section 6.8.4](#) for more details.
- **Final Certification** - Completed once the "Operational" status is achieved. Testing is to demonstrate that all environmental criteria are within project specifications. See [Section 6.8.4](#) for more details.
- **As-Built** - Condition where the cleanroom or clean zone is complete with all services connected and functioning but with no equipment, materials, or personnel present (ISO 14644-1:2015).
- **At-Rest** - Condition where cleanroom or clean zone is complete with equipment installed and operating in a manner agreed upon, but with no personnel present (ISO 14644-1:2015).
- **Operational** - Agreed condition where the cleanroom or clean zone is functioning in the specified manner, with equipment operating and with the specified number of personnel present (ISO 14644-1:2015).
- **FCT** - Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM.

- **GFTT** - Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards.
- **GFC** - Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities .
- **Global Facilities Central Team** - This is used interchangeably with Innovation & Integration Team.
- **Global Facilities Construction Team** - This is used interchangeably with Global Construction Team.
- **Global Facilities Operations Support Team** - This is used interchangeably with Global Facilities Operations Team.
- **Global Facilities Planning Team** - This is used interchangeably with Global Facilities Line Planning Team.

Note: Refer to [Figure 1](#): Construction/Certification Flow Diagram

4.0 Safety

4.1 Safety References

- [MTI Fab Safety Home Page](#)

4.2 Itemized Safety Requirements

- 4.2.1 Please refer to contractor provided safety training and documentation regarding construction PPE and access to construction zones not yet occupied/accepted by Micron.
- 4.2.2 For PPE and general safety requirements of the cleanroom under the control of Micron. Refer to the [Global Fab GE - Cleanroom Standard Operating Procedure](#).

5.0 General Cleanroom Protocol

5.1 The following general cleanroom protocol must be followed at all times at all stages of construction (1-5):

- No smoking or tobacco products are allowed in the construction or cleanroom areas at any time during construction or operation.
Note: Tobacco products are allowed only in designated smoking areas.
- Gum, candy, and food/drinks are not allowed in the cleanroom area with the exception of bottled water.

Note: Designated break areas will be set up outside of the clean room area.

Note: Use pipe end caps to contain cardboard core of tape.

Important: No duct tape is allowed in the cleanroom area.

- All doors are to remain closed except for material or personnel passage. Doors are only to remain open long enough to allow passage of material/personnel. Doors are to be returned to their normally closed position immediately following material or personnel passage.
- Toolbox wheels, electrical wire cart wheels and any other equipment that has wheels must have protective covers.

Important: Tape over the wheels in order to protect the floor from damage and particulates. Exceptions may be granted by site Contamination Control Team.

- A minimum number of doors entering the future cleanroom area will be designated as authorized entries.

Note: These doors will display signs stating the construction level and specific requirements as outlined below.

- All other doors entering the future cleanroom will be labeled with "Caution Do Not Enter" tape and signs stating: "Do Not Enter."
- Clean up of construction cuttings, dust, debris etc. must be a continuous operation to avoid particulate that may migrate into the cleanroom later, after construction is complete, and to avoid trapping dust and debris in the walls, ceilings, etc.
- Construction debris is to be kept to a minimum in all construction areas outside of the cleanroom construction zone.
- All equipment, piping, duct work etc. must be wiped down prior to installation using plenum wipes moistened with either 70/30 DI/IPA Cleaner or propanol.
- The use of brooms and dust mops is only allowed during Level 1 type construction is allowed.
- Beginning in Stage 2, vacuums (HEPA filter type) shall be used to remove all small debris and drywall dust from studs, beams, structural members and duct work. All cutting/grinding work will require HEPA equipped vacuums to capture debris. NO BROOMS/ DUST PANS ALLOWED beginning with Stage 2 construction.

Important: Cleanup of construction dust and debris must be a continuous operation to avoid trapping dust and debris in the walls, ceilings etc.

- Any time walls, ceilings, floors etc. are being assembled it is critical that every effort be made to ensure no dirt is trapped.

Example: When attaching wall panels to studs the subcontractor must always have someone cleaning in between the studs as they progress so contaminants do not become trapped.

6.0 Cleanroom Construction

6.1 Construction Stage Milestones

Construction stage milestones:

- **Stage 1** - Building Shell Enclosed.
- **Stage 2** - Clean Zone Perimeter Boundaries Enclosed.
- **Stage 3** - Cleanroom Walls, Floors, and Ceiling Assembly Completed.
- **Stage 4** - Ceiling ULPA/HEPA Filters Installed.
- **Stage 5** - Certification Testing.

Stage transition will be a decision based on the stage of construction and coordination between Contamination Control, Site Facilities Services, Construction Supervisor or Specialist, and the Project Coordinator.

6.2 STAGE 1- Rough Construction Stage

6.2.1 Stage 1 Construction Activities:

- Demolition/excavation.
- Concrete floor finished and cured.
- Equipment rooms enclosed with access openings.
- Exterior walls, ceiling, doors, and windows fully installed and completed.

6.2.2 Stage 1 Cleanroom Protocol:

- Continuous cleaning of the cleanroom construction zone is required.

6.2.3 Stage 1 Cleanroom Gowning:

- Normal street clothes may be worn throughout the cleanroom construction zone during the Stage 1 of construction.
- Refer to contractor provided safety documentation and training for guidance on safety attire / PPE (hardhats, safety glasses, safety shoes/ harness/etc.).

6.3 STAGE 2 - Establishing the Clean Zone

6.3.1 Stage 2 Construction Activities:

- Concrete floor changes and preparation complete.
- Concrete floor sealed and cured.
- Clean zone perimeter walls complete.
- Clean zone perimeter walls sealed, painted, and dried.

- Fire barriers complete.
- Upper zones of fire suppression system complete.
- Building structure air tight.
- Either permanent or temporary personnel gowning area(s) established for entry into cleanroom construction zone.
- Either permanent or temporary equipment staging room established for introduction of equipment into clean zone.

6.4 Stage 2 Cleanroom Protocol:

- Continuous cleaning of the cleanroom construction zone is required.
- Cleanroom approved, HEPA equipped vacuums are to be used exclusively. No brooms or dust-mops are to be used within the cleanroom construction zone.
- All material entering the cleanroom construction zone is to be thoroughly wiped with 70/30 DI/IPA cleaning solution.
- All future enclosed surfaces such as walls, waffle slab, etc. are to be thoroughly cleaned of any dust or debris prior to painting or sealing.

6.5 Stage 2 Cleanroom Gowning:

- Normal street clothes may be worn throughout the cleanroom construction area during the Stage 2 of construction.
- Shoe covers, hairnets, beard/mouth covers, and gloves required for entry into cleanroom construction zone.
- Refer to contractor provided safety documentation/training for guidance on safety PPE (hardhats, safety glasses, safety shoes/harness etc.).

6.6 STAGE 3 - Cleanroom Walls, Ceiling, and Floors Completed

6.6.1 Stage 3 Construction Activities:

- Raised floor complete and level.
- Laminate floor material laid and sealed.
- Protective temporary floor covers complete.
- Ceiling assembly complete and level.
- Perimeter clean zone glazing complete.
- Interior walls erected.
- Doors installed and operable.

- Fire suppression system installed and tested.
- TGM alarm system installed and tested.
- Permanent lighting complete and operable.
- Smoke detectors installed and tested.
- Recirculation ducting complete and tested. • Makeup air ductwork complete and tested.
- Control devices complete.
- Housekeeping vacuum system complete and tested.
- FFUS with ULPA filters installed with pre-filters. FFUs are not to be operating at this stage, only in place.

6.6.2 Stage 3 Cleanroom Protocol:

- Continuous cleaning of the cleanroom construction zone is required.
- All material entering the cleanroom construction zone is to be cleaned thoroughly with 70/30 DI/IPA cleaning solution.
- Cleanroom approved, HEPA equipped vacuums are to be used exclusively. No brooms or dust-mops are to be used within the cleanroom construction zone.

6.6.3 Stage 3 Cleanroom Gowning:

- Hairnets, beard/mouth covers, frocks, gloves, and shoe covers are required for entry into the cleanroom construction zone.

6.7 STAGE 4 - Initial Cleanroom Certification; As-Built Status Achieved

6.7.1 Stage 4 Construction Activities:

- Floor, ceiling, and wall changes complete.
- Protective covers removed from floor, ceiling, and walls.
- Fire suppression system charged and operable.
- TGM system operable and reporting.
- Smoke detectors operable.
- Recirculation air system operable and adjusted.
- Makeup air system operable and adjusted.
- Control devices operable and reporting.
- Safety systems operable.
- FFUs powered up and operating.
- Ceiling grid and HEPA scan.
- Permanent gowning rooms completed.

- All permanent cleanroom doors installed and operable at all levels.
- Contamination levels stable within project ISO specifications.
- Initial deep clean completed.

6.7.2 Stage 4 Cleanroom Protocol:

- Continuous cleaning of the cleanroom construction zone is required.
- Cleanroom approved, HEPA equipped vacuums are to be used exclusively. No brooms or dust-mops are to be used within the cleanroom construction zone.
- All material entering cleanroom construction zone is to be cleaned thoroughly with 70/30 DI/IPA cleaning solution.
- Beginning with Stage 4, cleanroom material restrictions must be followed. Materials restricted from entry to the cleanroom include:
 - Fiber ropes, straps, tie-downs, etc.
 - Non-cleanroom approved tapes.
 - Non-cleanroom approved adhesives, solvents, sealants, lubricants, etc.
- If the cleanroom is to be used for photo-lithography, ensure that any materials requested for entry to the cleanroom are approved for use in photo-lithography areas.
- Consult site Contamination Control with any questions regarding a material's approval status before allowing its use within the cleanroom construction zone.

6.7.3 Stage 4 Clean Zone Gowning

- Hairnets and shoe covers required for entry to permanent gowning rooms.
- One-piece cleanroom jumpsuit including hood and booties.
- Gloves.

6.7.4 The following activities must be completed, and Initial Certification must be signed off by site Contamination Control before advancing to the next stage of construction:

- HEPA/ceiling grid scan completed, with all required repairs and replacements completed.
- Airborne particle counts challenged to project ISO specifications.
- Surface particle counts challenged to project ISO specifications.

- Rough temperature and humidity stabilized.
- Floor conductivity tested and acceptable.
- Airflow volume and balance completed.
- See [Section 7.0](#) for testing criteria, guidelines, and responsible parties.

6.8 Stage 5 - Intermediate and Final Qualification of Cleanroom and Installation of Production Equipment; Operational Status Achieved.

6.8.1 Stage 5 Construction Activities:

- All control systems tested and calibrated.
- Final air system balance completed.
- Contamination levels stabilized to project specified ISO class.
- Final deep clean completed.
- Final cleanroom certification signed off by site Contamination Control.

6.8.2 Stage 5 Cleanroom Protocol:

- Cleanroom is now owned and occupied by local Micron Technology Inc. site. Standard cleanroom protocol is to be followed as outlined in [Global Fab GE - Cleanroom Standard Operating Procedure](#)

6.8.3 Stage 5 Cleanroom Gowning:

- Standard cleanroom gowning protocol as documented in [Global Fab GE - Cleanroom Standard Operating Procedure.](#)

6.8.4 The following activities must be completed, and Intermediate and Final Certifications must be signed off by site Contamination Control before acceptance of ownership of cleanroom:

- Intermediate certification is to include:
- Temperature and humidity stabilized to 70 F +/- 20 and 45% +/- 3% respectively.
- Airborne particle counts stabilized to project defined ISO class.
- Surface particle counts stabilized to project defined ISO class.
- Final certification is to include:
- Temperature and humidity stabilized to 70 F +/- 20 and 45% +/- 3% respectively.

- Airborne particle counts stabilized to project defined ISO class.
- Surface particle counts stabilized to project defined ISO class.
- Sound levels acceptable to site EHSS.
- Light levels acceptable to site EHSS.
- Vibration levels acceptable to site Facility Engineering/Design Group.
- EMI levels acceptable to site Facility Engineering/Design Group.
- FFU airflow balanced to 80 fpm +/- 20%.

7.0 Cleanroom Certification Testing Criteria

7.1 Airborne Particle Concentrations

7.1.1 Airborne particle concentrations are to be measured based on ISO 14644-1:2015 standards.

7.1.1.1 Minimum number of sample points are determined by:
 $NL=27(A/1000)$

7.1.1.2 Where NL is the minimum number of sample locations and A is the area to be certified in m².

7.1.1.3 Refer to ISO 14644-1:2015 Annex A for more detailed information regarding classification of air cleanliness by particle concentration.

7.1.2 The initial cleanroom certification will be held to (at minimum) ISO Class 5 standards at the 0.5um particle size regime, or specified project ISO class.

7.1.3 The intermediate and final cleanroom certifications will be held to the ISO Class standards which corresponds to the project specified requirements for the cleanroom.

7.2 Temperature and Humidity

7.2.1 Temperature and humidity are to be measured based on ISO 14644-3:2005 as outlined in sections B.8 and B.9 respectively.

7.2.1.1 Temperature Testing

- A. For temperature measurements, the cleanroom will be divided into a grid of equal areas with one test

point per grid square. At least two test points are to be measured within the cleanroom.

- B. Temperature probe position will be no closer than 30 cm to walls, ceiling, or floor; temperature probes for certification are preferred to be placed at, or near, production level.
- C. Refer to ISO 14644-3:2005 Section B.8 for more information regarding temperature testing.

7.2.1.2 Humidity Testing

- A. For humidity testing, at least one test location per humidity control zone is required.
- B. Humidity test should be made in conjunction with temperature testing.
- C. Refer to ISO 14644-3:2005 Section B.9 for more information regarding humidity testing.

7.3 Air Balance Testing

7.3.1 To be completed and signed off by site Facilities.

7.4 Surface Resistance Testing (Floor Conductivity)

7.4.1 There will be at minimum one test location per 5k square feet of cleanroom space.

7.4.2 Resistivity is to be within 1.0×10^5 to 1.0×10^9 at each location.

7.4.3 Refer to Section 12, Table 5 of [Micron Global Quality ESD control Program Standard](#).

7.5 HEPA Filter and Ceiling Grid Testing

7.5.1 An airborne particle counter with 0.1um sensitivity and HHIPA "pistol" attachment is to be used for HEPA/ceiling grid qualification.

7.5.2 Each HEPA will be scanned completely with overlapping passes and a total sampling time of one minute.

7.5.3 Scan is to include the entire HEPA membrane area as well as periphery seals.

7.5.4 If a deficiency is found, a full 1 cfm measurement is to be taken to determine the severity of the leak and the location is to be marked with colored tape for easy, visible reference.

7.5.5 Verify all repairs and replacements with the above methodology.

7.6 Sound and Light Testing

7.6.1 To be completed and signed off by site EHSS.

7.7 Vibration Testing

7.7.1 To be completed by site Facilities Engineering/Design group.

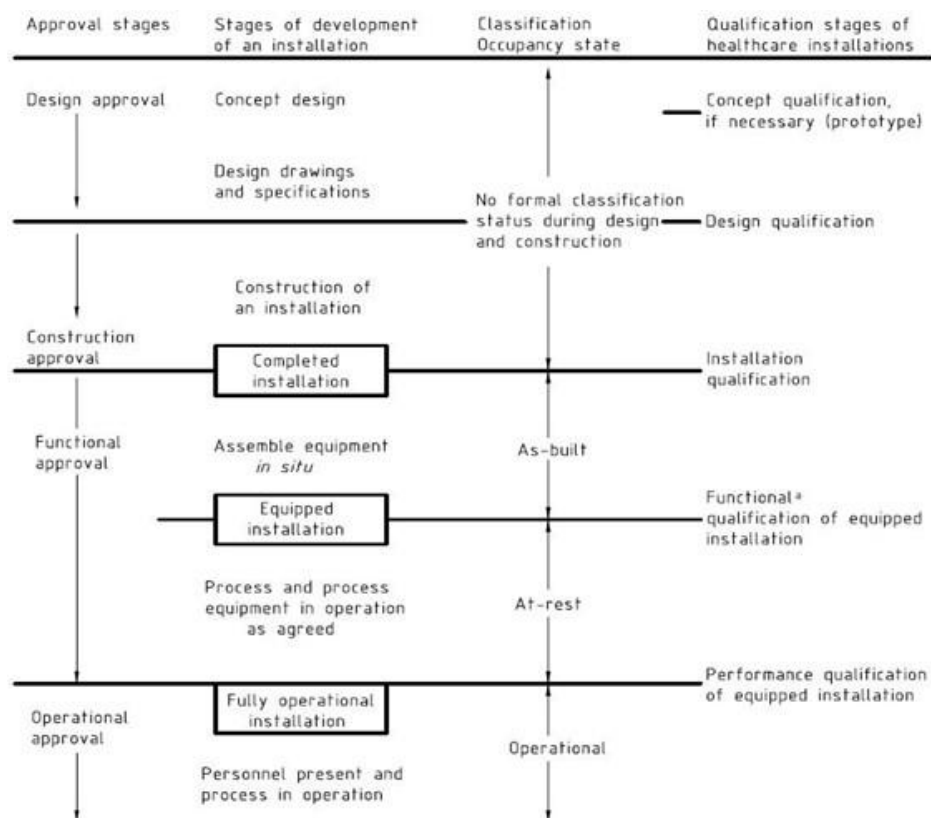
7.7.2 Vibration levels are to be no more than 125 μ in/second.

7.8 EMI Testing

7.8.1 To be completed by site Facilities Engineering/Design group.

7.8.2 EMI levels to be below 1mG at floor level.

Figure 1: Construction/Certification Flow Diagram: ISO 14644-4:2001 (Figure C.1; Page 22)



8.0 Document Control

8.1 Approval

FCG_ENG_CONSTRUCTION_APPR

8.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_NOTIFY
GLOBAL_FAC_CONST_SOP
GLOBAL_SECURITY_TEAM
GLOBAL_EHS_SAFETYHEALTH
GP_INDIRECT_FACILITIES
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
WW_FACILITIES_MANAGERS_APPR

8.3 Final Viewer Access

8.4 All Micron sites Facilities Team Members

8.5 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

8.6 Review

Biennial

9.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	N/A	New Document	07/25/19
1	8.0	Added Notification Group: GLOBAL_FAC_CONST_SOP	09/05/19
2	Overall document	General Formatting	10/01/21
	Title	Title changed from Global Facilities - Construction Cleanroom Protocol SOP to Global Facilities - Design & Construction - Construction - Cleanroom Protocol SOP	
	4.0	4.2.2: Link changed from F15 - Cleanroom Standard Operating Procedure to Global Fab GE - Cleanroom Standard Operating Procedure	
	6.0	6.8.2/6.8.3: Link changed from F15 - Cleanroom Standard Operating Procedure to Global Fab GE - Cleanroom Standard Operating Procedure	
	8.0	Approval updated to FCG_ENG_CONSTRUCTION_APPR from FCG_ENG_CONST_APPR Notification: See items in red for updates Was previously the following: WW_FACILITIES_MANAGERS, GFTT_LEADS, FPMO_PROJECT_MANAGERS, GLOBAL_FAC_CONST_SOP Final Viewer Access: See items in red for updates Was previously the following: All Facilities Departments Review: See items in red for updates	